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BS. Cyber Security

Network Security Project

**Simulated Office Network Security Using OPNsense Firewall and
Suricata IDS/IPS**

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Chapter 1: Introduction

1.1 Background

In modern enterprise environments, network security has become a critical concern as cyber threats continue to evolve in sophistication and frequency. Organizations of all sizes face threats including unauthorized access, port scanning, brute force attacks, Denial of Service (DoS) attacks, and malicious traffic targeting internal infrastructure. The deployment of firewalls and Intrusion Detection Systems (IDS) serves as the primary line of defense against such threats by monitoring, filtering, and logging network traffic based on predefined rules and behavioral signatures.

This project simulates a small office network environment using VMware virtualization, where a threat actor machine (Kali Linux) is used to perform common attack patterns against a legitimate Windows 10 client. The network is protected by OPNsense, an open-source firewall and routing platform, supplemented by Suricata, a high-performance network IDS/IPS engine. This setup closely mirrors real-world enterprise security deployments and demonstrates practical skills in network defense.

1.2 Problem Statement

Small office and enterprise networks are increasingly vulnerable to internal threats, misconfigured services, and lateral movement by attackers who have gained access to the internal network. Without proper traffic filtering and intrusion detection, a compromised internal machine can perform unrestricted port scanning, brute force credential attacks on services like SSH, flood systems with DoS traffic, and exfiltrate data over common ports such as HTTP and HTTPS.

This project addresses the challenge of securing an internal LAN segment against such threats using a multi-layer security approach that combines stateful firewall rule enforcement, real-time intrusion detection via Suricata, live traffic monitoring, and scalable rule management through Aliases.

1.3 Objectives

- Deploy and configure OPNsense as a network gateway and firewall on a simulated LAN.
- Create specific firewall rules to block malicious traffic from Kali Linux to Windows 10.
- Install and configure Suricata IDS in PCAP mode on the LAN interface.
- Enable ET Open rulesets and create custom user-defined rules to detect Kali Linux traffic.
- Monitor and verify all blocked and allowed traffic using OPNsense Live View.
- Test firewall rules by actively generating attack traffic from Kali Linux and confirming blocks.
- Demonstrate brute force detection using Suricata alerts.
- Implement Aliases for scalable and maintainable firewall rule management.

1.4 Scope

This project is confined to a simulated virtualized network environment. It does not represent a live production deployment. The scope covers the following:

- A three-node virtual network: OPNsense Firewall, Kali Linux (attacker), and Windows 10 (victim/client).
- Firewall rule implementation and testing for Layer 3/4 traffic filtering.
- Suricata IDS configuration for scan, DoS, brute force, and web attack detection.
- Real-time traffic log monitoring and analysis.
- Dynamic Alias-based rule management demonstrating scalability.

The project does not cover wireless security, application-layer deep packet inspection beyond Suricata signatures, VPN configuration, or external WAN-side attack simulation.

Chapter 2: Tools and Technologies

2.1 OPNsense

OPNsense is a free, open-source firewall and routing platform based on FreeBSD. It is derived from pfSense and m0n0wall and offers a modern, intuitive web-based interface for managing network security. OPNsense supports stateful packet inspection, NAT, VPN, traffic shaping, and an extensive plugin system. In this project, OPNsense serves as the gateway firewall at IP address 10.0.0.2, enforcing all LAN-side firewall rules and hosting the Suricata IDS/IPS service. Key features used include the Firewall Rules engine, Live Log View, Aliases, and the Intrusion Detection (Suricata) plugin.

2.2 VirtualBox / VMware

The network environment was built using a virtualization platform (VMware/VirtualBox) to create isolated virtual machines on a shared host. Virtualization allows a controlled, reproducible environment where network interfaces, IP addressing, and inter-VM communication can be precisely configured. All three machines (OPNsense, Kali Linux, Windows 10) share a Host-Only or Internal network adapter, placing them on the same simulated LAN segment (10.0.0.0/24).

2.3 Kali Linux

Kali Linux is a Debian-based Linux distribution specifically designed for penetration testing and offensive security research. In this project, it functions as the simulated threat actor at IP address 10.0.0.73. Attack tools used include nmap (port scanning), hydra (SSH brute force), and standard ping/ICMP utilities. Kali Linux represents a compromised internal machine attempting lateral movement and reconnaissance within the office network.

2.4 Windows 10

Windows 10 at IP address 10.0.0.62 represents the legitimate internal client or workstation in the simulated office environment. It is the primary target of attacks originating from the Kali Linux machine. The firewall rules are designed to protect this host from direct access by the Kali

machine while allowing it full unrestricted outbound access to the network and internet.

2.5 Suricata IDS/IPS

Suricata is a high-performance, open-source Network Intrusion Detection System (NIDS), Intrusion Prevention System (IPS), and Network Security Monitoring (NSM) engine. It is maintained by the Open Information Security Foundation (OISF). In this project, Suricata is deployed in PCAP live mode on the LAN interface of OPNsense, passively inspecting all traffic and generating alerts based on signature matches. The ET Open (Emerging Threats) ruleset provides community-maintained signatures for detecting DoS, scanning, brute force, and web attack patterns. Additionally, a custom user-defined rule was created to detect and alert on any traffic originating from the Kali Linux IP address.

Chapter 3: System Architecture

3.1 Network Topology

The simulated network consists of three virtual machines interconnected on a single LAN segment (10.0.0.0/24). OPNsense acts as the gateway/router and firewall, connected to both the simulated LAN and a WAN interface for external access. Kali Linux and Windows 10 are connected solely to the LAN segment and communicate through the OPNsense firewall.

3.2 IP Configuration

Machine	IP Address	Role
OPNsense Firewall	10.0.0.2	Gateway / Firewall / IDS Host
Kali Linux	10.0.0.73	Simulated Threat Actor (Attacker)
Windows 10	10.0.0.62	Legitimate Internal Client (Target)

3.3 Workflow

The security workflow operates in four layers:

- Layer 1 — Firewall Rules: OPNsense inspects all incoming LAN traffic and applies stateful firewall rules. Matching packets from Kali Linux are immediately blocked (Quick match) and logged. Packets from Windows 10 are passed with full access.
- Layer 2 — Suricata IDS: Suricata runs passively on the LAN interface in PCAP mode, matching traffic against ET Open rulesets and a custom user-defined rule. When signatures match, alerts are generated and stored in the Suricata alert log with timestamp, SID, source/destination, and alert message.
- Layer 3 — Traffic Monitoring: The OPNsense Firewall Live View provides real-time visibility into all matched traffic, showing whether each connection was blocked or passed and which rule applied.
- Layer 4 — Alias Management: Aliases abstract IP addresses and port groups into named objects that are referenced by firewall rules. Updating an alias automatically propagates the change to all rules that use it, enabling scalable and maintainable rule management.

Chapter 4: Implementation

Task 1: Firewall Rules

In this task, all firewall rules were configured on the LAN interface of OPNsense to control traffic originating from Kali Linux (10.0.0.73) and Windows 10 (10.0.0.62). Rules are processed in top-down order with Quick match enabled, meaning the first matching rule is applied immediately and processing stops.

Overview of All Configured Rules

The screenshot below shows the complete set of firewall rules configured on the LAN interface. Rules with a red X icon are blocking rules; rules with a green play icon are pass rules. The rules are evaluated in order from top to bottom.

Firewall: Rules: LAN

Select category

Inspect

The changes have been applied successfully.

☐	Protocol	Source	Port	Destination	Port	Gateway	Schedule		Description						
	Automatically generated rules											23			
☐		IPv4 *	kali	*	10.0.0.62/32	*	*	*	Rule 1 - Block Kali to Windows						
☐		IPv4 ICMP	kali	*	*	*	*	*	Rule 4 - Block ICMP from Kali						
☐		IPv4 TCP	kali	*	*	SSH (22)	*	*	Rule 2 - Block SSH from Kali						
☐		IPv4 *	kali	*	*	*	*	*	Rule 5 - Log and Block all remaining Kali traffic						
☐		IPv4 *	10.0.0.62/32	*	*	*	*	*	Rule 3 - Allow Windows Full Access						
☐		IPv4 *	LAN net	*	*	*	*	*	Default allow LAN to any rule						
☐		IPv6 *	LAN net	*	*	*	*	*	Default allow LAN IPv6 to any rule						
	pass		block		reject		log		in		first match				
	pass (disabled)		block (disabled)		reject (disabled)		log (disabled)		out		last match				
	Active/Inactive Schedule (click to view/edit)														
	Alias (click to view/edit)														

Figure 1.1 — OPNsense LAN Firewall Rules Overview showing all configured rules

The rule set consists of six custom rules plus the default LAN pass rules. Rules 1 through 3 and Rules 5 and 6 block various categories of Kali Linux traffic. Rule 4 explicitly allows full access for the Windows 10 machine. The default LAN and LAN IPv6 rules at the bottom permit all other traffic.

Rule 1 — Block Kali to Windows (All Traffic)

This rule blocks all IPv4 traffic from the Kali Linux machine (10.0.0.73/32) destined for the Windows 10 machine (10.0.0.62/32). The action is set to Block with Quick match enabled, ensuring any direct communication attempt from Kali to Windows is immediately dropped. The Log option is enabled so that all matched packets are recorded in the firewall log for audit purposes.

The screenshot shows the 'Firewall: Rules: LAN' configuration page. The 'Edit Firewall rule' section is active. The configuration is as follows:

Field	Value
Action	Block
Disabled	☐ Disable this rule
Quick	☒ Apply the action immediately on match.
Interface	LAN
Direction	in
TCP/IP Version	IPv4
Protocol	any

Figure 1.2 — Rule 1 configuration: Action set to Block, Interface LAN, Protocol any

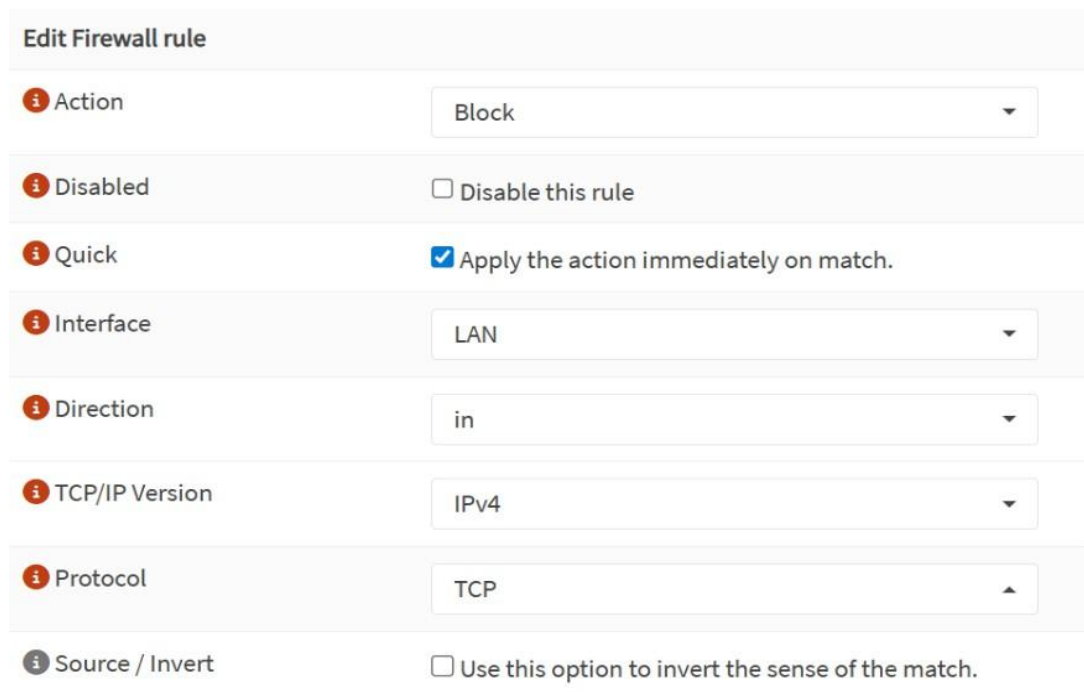
The screenshot shows the 'Advanced' tab of the Firewall rule configuration. The settings are as follows:

Field	Value
Source / Invert	☐ Use this option to invert the sense of the match.
Source	Single host or Network 10.0.0.73/32
Destination / Invert	☐ Use this option to invert the sense of the match.
Destination	Single host or Network 10.0.0.62/32
Destination port range	from: any to: any
Log	☒ Log packets that are handled by this rule

Figure 1.3 — Rule 1 source/destination: Source 10.0.0.73/32, Destination 10.0.0.62/32, Logging enabled

Rule 2 — Block SSH from Kali (Port 22)

This rule specifically blocks TCP traffic from the Kali Linux machine targeting port 22 (SSH) on any destination. The protocol is set to TCP and the destination port range is locked to SSH (22). This prevents Kali from performing SSH brute force attacks against any host on the network, not only Windows 10. Logging is enabled to capture all SSH connection attempts originating from the threat actor.



Edit Firewall rule

Action Block

Disabled ☐ Disable this rule

Quick ☒ Apply the action immediately on match.

Interface LAN

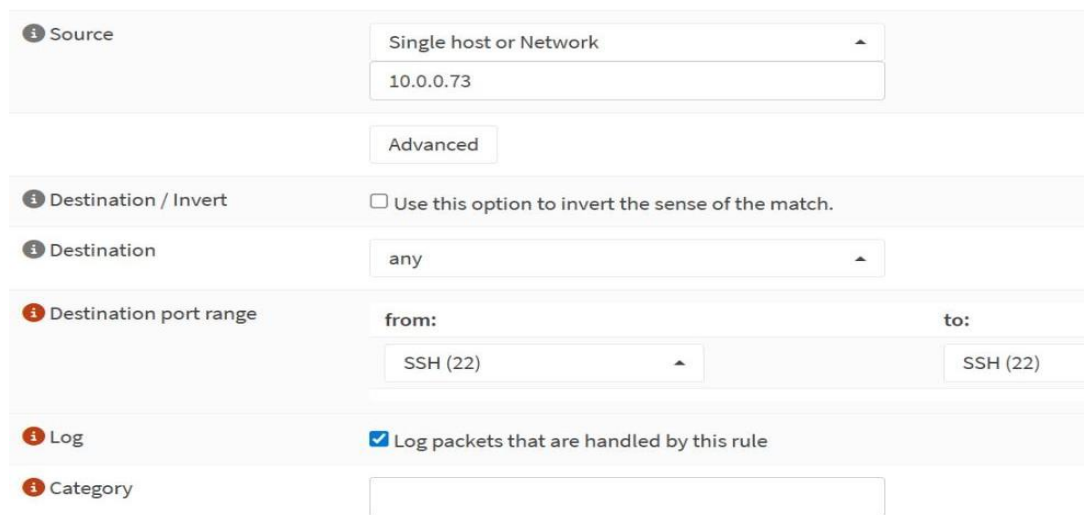
Direction in

TCP/IP Version IPv4

Protocol TCP

Source / Invert ☐ Use this option to invert the sense of the match.

Figure 1.4 — Rule 2 configuration: Action Block, Protocol TCP, targeting SSH



Source Single host or Network
10.0.0.73

Advanced

Destination / Invert ☐ Use this option to invert the sense of the match.

Destination any

Destination port range from: SSH (22) to: SSH (22)

Log ☒ Log packets that are handled by this rule

Category

Figure 1.5 — Rule 2 source/destination port: Source 10.0.0.73, Destination port SSH (22), Logging enabled

Rule 3 — Allow Windows Full Access

This is a Pass rule that explicitly permits all IPv4 traffic from the Windows 10 machine (10.0.0.62) to any destination without restriction. By placing this rule after the Kali blocking rules but before the default rules, it ensures that the legitimate client machine retains full network access. The Quick match flag is enabled so that once matched on 10.0.0.62, no further rules are evaluated for Windows traffic.

Edit Firewall rule

Action	Pass
Disabled	☐ Disable this rule
Quick	☒ Apply the action immediately on match.
Interface	LAN
Direction	in
TCP/IP Version	IPv4
Protocol	any
Source / Invert	☐ Use this option to invert the sense of the match.

Figure 1.6 — Rule 3 configuration: Action Pass, Interface LAN, Protocol any

Source / Invert ☐ Use this option to invert the sense of the match.

Source Single host or Network
10.0.0.62

Advanced

Destination / Invert ☐ Use this option to invert the sense of the match.

Destination any

Destination port range from: any to: any

Log ☒ Log packets that are handled by this rule

Category

Figure 1.7 — Rule 3 source: 10.0.0.62, Destination any, Logging enabled

Rule 4 — Block ICMP from Kali

This rule blocks ICMP traffic (ping) from the Kali Linux machine. Setting the Protocol to ICMP with ICMP type set to Any prevents all forms of ICMP communication from the attacker, including ping sweeps, traceroute probes, and ICMP-based DoS attacks. This effectively makes the Kali machine unable to ping or use ICMP-based reconnaissance tools against any network host.

Edit Firewall rule

Action Block

Disabled ☐ Disable this rule

Quick ☒ Apply the action immediately on match.

Interface LAN

Direction in

TCP/IP Version IPv4

Protocol ICMP

ICMP type Any

Figure 1.8 — Rule 4 configuration: Action Block, Protocol ICMP, ICMP type Any

Source / Invert ☐ Use this option to invert the sense of the match.

Source Single host or Network
10.0.0.73

Advanced

Destination / Invert ☐ Use this option to invert the sense of the match.

Destination any

Destination port range from: any to: any

Log ☒ Log packets that are handled by this rule

Figure 1.9 — Rule 4 source: 10.0.0.73, Destination any, Logging enabled

Rule 5 — Log and Block All Remaining Kali Traffic

This is a catch-all rule that blocks and logs any remaining IPv4 traffic from 10.0.0.73 that was not matched by earlier specific rules. It ensures that no traffic from the Kali machine can pass through the firewall regardless of port or protocol. The description field confirms the intent: "Rule 5 - Log and Block all remaining Kali traffic". This acts as a final safety net in the rule chain.

Edit Firewall rule

Action Block

Disabled ☐ Disable this rule

Quick ☒ Apply the action immediately on match.

Interface LAN

Direction in

TCP/IP Version IPv4

Protocol any

Source / Invert ☐ Use this option to invert the sense of the match.

Figure 1.10 — Rule 5 configuration: Action Block, Protocol any, Source 10.0.0.73

Description Rule 6 - Log and Block all remaining Kali traffic

No XMLRPC Sync ☐

Schedule none

Gateway default

Traffic shaping

rule direction	reverse direction
None	None

Advanced features Show/Hide

Save **Cancel**

Figure 1.11 — Rule 5 description and save confirmation showing the catch-all block rule

Task 2: Suricata IDS/IPS

Suricata was deployed as an Intrusion Detection System (IDS) on the OPNsense firewall under Services > Intrusion Detection. It operates in PCAP live mode on the LAN interface, passively monitoring all traffic and generating alerts when signatures match. The configuration involved downloading rulesets, defining detection policies, reviewing active rules, creating a custom user-defined rule, and verifying alert generation.

Ruleset Download

The Download tab in the Suricata administration panel was used to select and enable the Emerging Threats (ET) Open rulesets. From the available list, the following rulesets were enabled and downloaded: ET open/emerging-scan for port scan detection, and ET open/emerging-worm for network worm detection, and ET open/emerging-dos for DOS attacks detection and ET open/tor for suspicious traffic detection. Other available rulesets such as emerging-dos, emerging-shellcode, and emerging-smtp are visible but not installed. The "Download and Update Rules" button was used to fetch the latest rule definitions.

☐	ET open/emerging-scan	2026/05/13 9:47	✓	
☐	ET open/emerging-worm	2026/05/13 9:47	✓	
☐	ET open/threatview_CS_c2	not installed	×	
☐	ET open/tor	2026/05/13 9:47	✓	
☐	ET open/emerging-dos	2026/05/13 9:47	✓	

Figure 2.1 — Suricata Download tab showing ET Open rulesets

Detection Policies

Three detection policies were configured under Services > Intrusion Detection > Policy. Each policy maps a specific ruleset to a detection action:

- DoS Attack Detection uses emerging-dos.rules with Alert action to detect Denial of Service attack patterns.
- Kali Scan Detection Policy uses emerging-scan.rules with Alert action to detect port scan and network reconnaissance behavior originating from the Kali machine.

- Web Application Attack Detection Policy uses tor.rules with Alert action to detect Tor-based web application attack patterns.

All three policies are enabled (checkmark in Enabled column) with Priority 1 and both the original Action and New action set to Alert. This configuration ensures that matching traffic generates alerts without actively blocking it (IDS mode rather than IPS mode).

Services: Intrusion Detection: Policy

Policies							
Rule adjustments							
☐	Enabled	Priority	Rulesets	Action	New action	Description	Commands
☐	☒	1	emerging-dos.rules	Alert	Alert	DoS Attack Detection	  
☐	☒	1	emerging-scan.rules	Alert	Alert	Kali Scan Detection Policy	  
☐	☒	1	tor.rules	Alert	Alert	Web Application Attack Detection Policy	  



1



Apply

Figure 2.2 — Suricata Policy tab showing three active detection policies for DoS, Scan, and Web Attack detection

Active Rules

The Rules tab displays all individual Suricata rules that are currently active. The list shows rules from the emerging-dos and emerging-scan rulesets with SIDs such as 2000006, 2000010, 2000011, 2000536, 2000537, 2000538, 2000540, 2000543, and others. All visible rules have logging enabled (checkmark in Info/Enabled column). The pagination shows five pages of active rules, indicating a comprehensive rule set is loaded.

Settings	Download	Rules	User defined	Alerts	Schedule
Filters					
Search 50 Alert Drop					
☐ sid	Action	Source	ClassType	Message	Info / Enabled
☐ 4294967293	drop	OPNsens...	##none##	BruteForce Block	☒
☐ 2020702	alert	emergen...	attempt...	ET DOS Bittorrent User-Agent inbound - possible DDOS	☒
☐ 2024584	alert	emergen...	attempt...	ET DOS CLDAP Amplification Reflection (PoC based)	☒
☐ 2030339	alert	emergen...	attempt...	ET DOS CallStranger - Attempted UPnP Reflected Amplified TCP with Multip...	☒
☐ 2000011	alert	emergen...	attempt...	ET DOS Catalyst memory leak attack	☒
Showing 1 to 50 of 1400 entries					

Figure 2.3 — Suricata Rules tab showing active ET Open rules for DoS and Scan detection

General Settings

The Settings tab confirms the core Suricata configuration. The service is Enabled, Capture mode is set to IPS mode, and Promiscuous mode is enabled to capture all traffic on the interface regardless of destination MAC address. The Interfaces field shows LAN, confirming monitoring is performed on the internal network interface. Under Logging, Enable syslog alerts is checked, Rotate log is set to Weekly, and Save logs is set to retain 4 log files.

The Pattern matcher is set to Default.

Settings	Download	Rules	User defined	Alerts	Schedule
☒ advanced mode full help					
General Settings					
Enabled		☒			
Capture mode		PCAP live mode (IDS)			
Promiscuous mode		☒			
Interfaces		LAN			
		☒ Clear All ☒ Select All			
Detection					
Pattern matcher		Default			
Logging					
Enable syslog alerts		☒			
Enable eve syslog output		☐			
Rotate log		Weekly			
Save logs		4			

Figure 2.4 — Suricata Settings tab: PCAP live mode (IDS), LAN interface, syslog alerts enabled

User-Defined Rule

A custom user-defined rule was created under the User Defined tab to explicitly detect and alert on all traffic from the Kali Linux machine. The rule is enabled, the Action is set to Alert, and the Description reads "Kali Linux Traffic Detection". This rule complements the ET Open signatures by providing a direct, purpose-built alert for any packet whose source matches the Kali IP address, ensuring no Kali traffic goes undetected even if it does not match standard signatures.

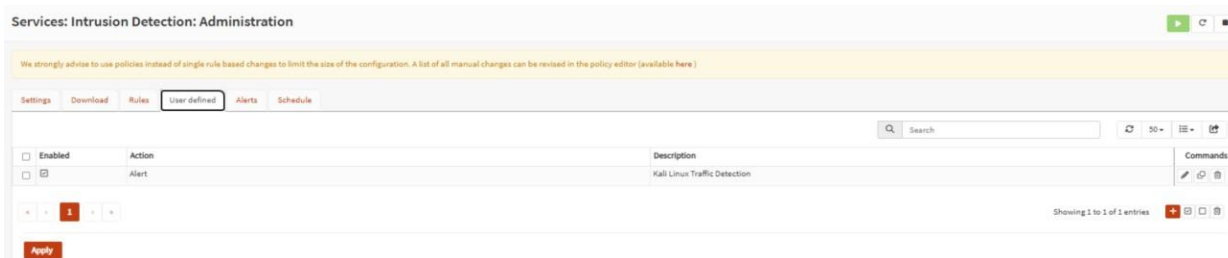


Figure 2.5 — Suricata User Defined tab showing the custom "Kali Linux Traffic Detection" alert rule

IDS Alerts — Custom Rule Triggered

The Alerts tab shows the output of the custom user-defined rule after Kali traffic was generated. Multiple alert entries are visible, all timestamped on 2026-05-17 at 09:34 to 09:33. Each entry shows Source 10.0.0.73 with a high ephemeral source port, Destination 10.0.0.2 on port 53 (DNS), Action allowed (IDS mode, not blocking), Interface LAN, and Alert message "Kali Linux scan traffic". This confirms that the custom rule successfully detected and logged DNS-based scan queries from the Kali machine to the OPNsense firewall itself. A total of 11 entries were recorded.

The screenshot shows the OPNsense web interface. On the left is a sidebar menu with options like Reporting, System, Interfaces, Firewall, VPN, Services, Captive Portal, DHCP Relay, Dnsmasq DNS & DHCP, Intrusion Detection, Administration, Policy, Log File, Kea DHCP, Monit, Network Time, OpenDNS, Router Advertisements, Unbound DNS, Power, and Help. The main content area is titled 'Services: Intrusion Detection: Administration'. It features a yellow warning banner at the top and a navigation bar with tabs: Settings, Download, Rules, User defined, Alerts, and Schedule. The 'Alerts' tab is active, displaying a table of alerts. The table has columns for Timestamp, SID, Action, Interface, Source, Port, Destination, Port, Alert, and Info. The alerts listed are all 'allowed' actions on the 'LAN' interface, triggered by 'Kali Linux scan traffic' from source IP 10.0.0.73 to various destination ports (59031, 48219, 59982, 52203, 54300, 38907). The interface also includes a search bar, a date filter set to '2026/05/17 9:23', and a pagination control showing 'Showing 1 to 11 of 11 entries'.

Timestamp	SID	Action	Interface	Source	Port	Destination	Port	Alert	Info
2026-05-17T09:34:15.094325+0000	429496...	allowed	LAN	10.0.0.73	59031	10.0.0.2	53	Kali Linux scan traffic	
2026-05-17T09:33:20.381622+0000	429496...	allowed	LAN	10.0.0.73	48219	10.0.0.2	53	Kali Linux scan traffic	
2026-05-17T09:33:18.615368+0000	429496...	allowed	LAN	10.0.0.73	59982	10.0.0.2	53	Kali Linux scan traffic	
2026-05-17T09:33:18.598351+0000	429496...	allowed	LAN	10.0.0.73	52203	10.0.0.2	53	Kali Linux scan traffic	
2026-05-17T09:33:18.595065+0000	429496...	allowed	LAN	10.0.0.73	54300	10.0.0.2	53	Kali Linux scan traffic	
2026-05-17T09:33:18.582666+0000	429496...	allowed	LAN	10.0.0.73	38907	10.0.0.2	53	Kali Linux scan traffic	

Figure 2.6 — Suricata Alerts showing custom rule "Kali Linux scan traffic" triggered by Kali DNS queries

IDS Alerts — Kali to Windows Scan Traffic Detected

A second set of IDS alerts shows the custom rule detecting Kali traffic directed at the Windows 10 machine. All entries are timestamped at 2026-05-17 09:45, with Source 10.0.0.73, Destination 10.0.0.62, and varying high-numbered destination ports (1666, 666, 16001, 417, 27715, 6123 etc.), confirming port scanning behavior. The Alert message reads "Kali Traffic To Windows Detected". A total of 51 entries are recorded across two pages, demonstrating that the scan was extensive and that Suricata successfully captured each connection attempt despite the firewall having already blocked them at Layer 1.

```
(haleema@Haleema)-[~]
$ sudo nmap -sS 10.0.0.62
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-17 05:45 EDT
Nmap scan report for 10.0.0.62
Host is up (0.31s latency).
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
5357/tcp  open  wsddapi
MAC Address: 00:0C:29:BE:C0:E3 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 2.15 seconds
```

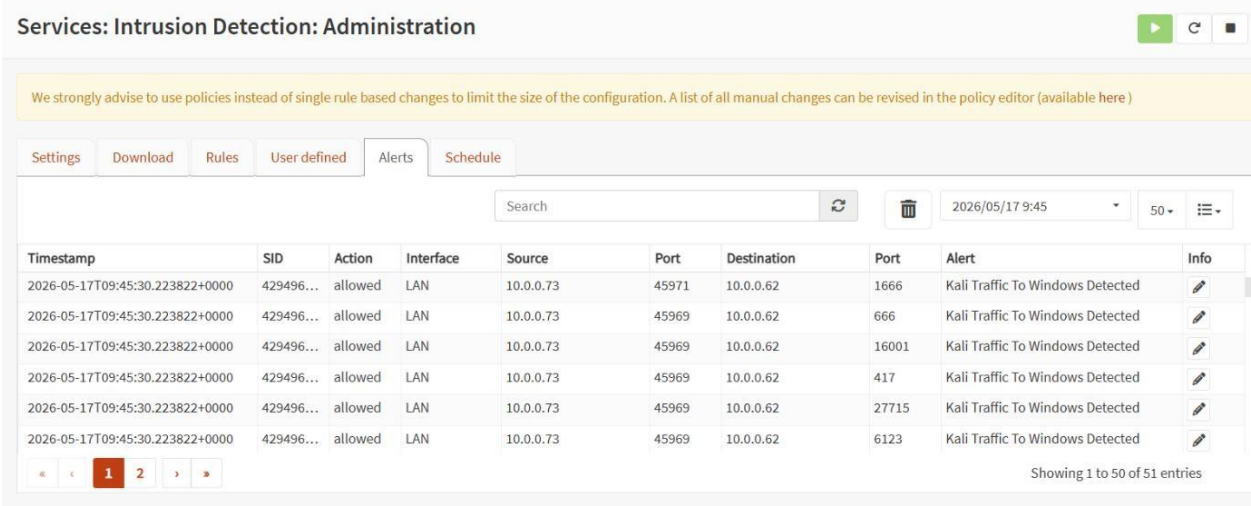


Figure 2.7 — Suricata Alerts showing "Kali Traffic To Windows Detected" from the custom rule across 51 entries

Task 3: Traffic Monitoring

Real-time traffic monitoring was performed using the OPNsense Firewall Live View (Firewall > Log Files > Live View). This tool provides a live stream of all firewall-matched events with columns for Interface, Direction, Time, Protocol, Source, Destination, Action, and the matching Rule Label. Rows are color-coded: red for blocked traffic and green for passed traffic.

Kali Linux Traffic — Blocked

With a filter applied for source containing "10.0.0.73", the Live View clearly shows all Kali traffic being blocked in real time. The highlighted red rows show UDP packets from 10.0.0.73 to 10.0.0.2:53 (DNS) being blocked by Rule 6 (Log and Block all remaining Kali traffic), as well as ICMP packets from 10.0.0.73 to 8.8.8.8 being blocked by Rule 5 (Block ICMP from Kali). This confirms that the firewall is actively enforcing both the specific ICMP block rule and the catch-all block rule against all Kali-originating traffic.

Firewall: Log Files: Live View

The screenshot shows the OPNsense Firewall Live View interface. At the top, there are filters and options. The filter is set to 'src contains 10.0.0.73'. The options include 'Auto-refresh' (checked), 'Lookup hostnames' (unchecked), and 'Select any of given criteria (or)' (unchecked). The table size is set to 25 and the history size is 10000. The table below shows the following data:

Interface	Time	Protocol	Source	Destination	Action	Label
LAN	In 2026-05-17T10:58:45	UDP	10.0.0.73:46848	10.0.0.2:53	block	Rule 6 - Log and Block all remaining Kali traffic
LAN	In 2026-05-17T10:58:40	UDP	10.0.0.73:52423	10.0.0.2:53	block	Rule 6 - Log and Block all remaining Kali traffic
LAN	In 2026-05-17T10:58:40	UDP	10.0.0.73:52423	10.0.0.2:53	block	Rule 6 - Log and Block all remaining Kali traffic
LAN	In 2026-05-17T10:58:35	UDP	10.0.0.73:52423	10.0.0.2:53	block	Rule 6 - Log and Block all remaining Kali traffic
LAN	In 2026-05-17T10:58:35	UDP	10.0.0.73:52423	10.0.0.2:53	block	Rule 6 - Log and Block all remaining Kali traffic
LAN	In 2026-05-17T10:53:11	ICMP	10.0.0.73	8.8.8.8	block	Rule 5 - Block ICMP from Kali
LAN	In 2026-05-17T10:53:10	ICMP	10.0.0.73	8.8.8.8	block	Rule 5 - Block ICMP from Kali
LAN	In 2026-05-17T10:53:09	ICMP	10.0.0.73	8.8.8.8	block	Rule 5 - Block ICMP from Kali

Figure 3.1 — Live View filtered by src=10.0.0.73 showing all Kali traffic blocked (red rows), matched against Rule 5 and Rule 6

Non-Kali Traffic — Passed

With the filter set to "src does not contain 10.0.0.73", the Live View shows all other network traffic. The green rows represent outbound WAN traffic from the OPNsense firewall itself (192.168.1.19 range to external IPs) passing through the "let out anything from firewall host itself (force gw)" rule. This confirms that legitimate traffic from the firewall and other sources is flowing normally, while only Kali traffic is being blocked.

Firewall: Log Files: Live View

Filters

Quick search (all fields)...

src does not contain 10.0.0.73 Apply

Active filters: src does not contain "10.0.0.73" x

Templates

Choose template

Options

- ☒ Auto-refresh
- ☐ Lookup hostnames
- ☐ Select any of given criteria (or)

25 Table size

10000 History size

Interface		Time	Protocol	Source	Destination	Action	Label	
WAN	Out	2026-05-17T11:05:22	UDP	192.168.1.19:53...	23.74.25.192:53	pass	let out anything from firewall host itself (force gw)	1
WAN	Out	2026-05-17T11:05:22	UDP	192.168.1.19:4866	23.74.25.192:53	pass	let out anything from firewall host itself (force gw)	1
WAN	Out	2026-05-17T11:05:22	UDP	192.168.1.19:51...	193.108.88.1:53	pass	let out anything from firewall host itself (force gw)	1
WAN	Out	2026-05-17T11:05:22	UDP	192.168.1.19:24...	23.74.25.192:53	pass	let out anything from firewall host itself (force gw)	1
WAN	Out	2026-05-17T11:05:22	UDP	192.168.1.19:14...	84.53.139.193:53	pass	let out anything from firewall host itself (force gw)	1
WAN	Out	2026-05-17T11:05:22	UDP	192.168.1.19:39...	23.61.199.193:53	pass	let out anything from firewall host itself (force gw)	1
WAN	Out	2026-05-17T11:05:21	UDP	192.168.1.19:30...	193.108.88.1:53	pass	let out anything from firewall host itself (force gw)	1
WAN	Out	2026-05-17T11:05:21	UDP	192.168.1.19:14...	193.108.88.1:53	pass	let out anything from firewall host itself (force gw)	1

Figure 3.2 — Live View filtered to exclude Kali traffic showing all other traffic passing normally (green rows)

Rule 3 — SSH Block Verification in Live View

Filtering the Live View by label containing "Rule 3" isolates all entries matching the SSH blocking rule. The red rows show multiple TCP connections from 10.0.0.73 on ephemeral source ports (57222, 35812) targeting 10.0.0.62:22 (SSH) being blocked at various timestamps. Multiple repeated attempts from the same source port within one second confirm SSH brute force attempt behavior, all of which are being successfully blocked by Rule 3.

Firewall: Log Files: Live View

Filters

Quick search (all fields)...

label contains Rule 3 Apply

Active filters: label contains "Rule 3" x

Templates

Choose template

Options

- ☒ Auto-refresh
- ☐ Lookup hostnames
- ☐ Select any of given criteria (or)

25 Table size

10000 History size

Interface		Time	Protocol	Source	Destination	Action	Label	
LAN	In	2026-05-17T10:13:35	TCP	10.0.0.73:57222	10.0.0.62:22	block	Rule 3 - Block SSH from Kali	1
LAN	In	2026-05-17T10:13:31	TCP	10.0.0.73:57222	10.0.0.62:22	block	Rule 3 - Block SSH from Kali	1
LAN	In	2026-05-17T10:13:31	TCP	10.0.0.73:57222	10.0.0.62:22	block	Rule 3 - Block SSH from Kali	1
LAN	In	2026-05-17T10:13:31	TCP	10.0.0.73:57222	10.0.0.62:22	block	Rule 3 - Block SSH from Kali	1
LAN	In	2026-05-17T08:26:59	TCP	10.0.0.73:35812	10.0.0.62:22	block	Rule 3 - Block SSH from Kali	1
LAN	In	2026-05-17T08:26:59	TCP	10.0.0.73:35812	10.0.0.62:22	block	Rule 3 - Block SSH from Kali	1
LAN	In	2026-05-17T08:26:59	TCP	10.0.0.73:35812	10.0.0.62:22	block	Rule 3 - Block SSH from Kali	1
LAN	In	2026-05-17T08:26:59	TCP	10.0.0.73:35812	10.0.0.62:22	block	Rule 3 - Block SSH from Kali	1

Figure 3.3 — Live View filtered by "Rule 3" showing repeated SSH connection attempts from Kali blocked in real time

Rule 4 — Windows Full Access Verification in Live View

Filtering by label containing "Rule 4" shows all Windows 10 traffic being passed. The green rows display TCP and UDP connections from 10.0.0.62 to various external destinations (103.99.132.127:80, 13.79.163.91:443, 110.93.229.202:80, 10.0.0.2:53) all being permitted under Rule 4. This confirms that despite the extensive blocking applied to Kali traffic, the Windows machine retains complete and unrestricted network access.

Firewall: Log Files: Live View

Interface	Time	Protocol	Source	Destination	Action	Label
LAN	In 2026-05-17T11:02:49	TCP	10.0.0.62:49646	103.99.132.127:80	pass	Rule 4 - Allow Windows Full Access
LAN	In 2026-05-17T11:02:43	TCP	10.0.0.62:49645	103.99.132.127:80	pass	Rule 4 - Allow Windows Full Access
LAN	In 2026-05-17T11:02:42	TCP	10.0.0.62:49644	13.79.163.91:443	pass	Rule 4 - Allow Windows Full Access
LAN	In 2026-05-17T11:02:33	TCP	10.0.0.62:49643	103.99.132.127:80	pass	Rule 4 - Allow Windows Full Access
LAN	In 2026-05-17T11:02:19	TCP	10.0.0.62:49642	13.79.163.91:443	pass	Rule 4 - Allow Windows Full Access
LAN	In 2026-05-17T11:02:17	TCP	10.0.0.62:49641	110.93.229.202:80	pass	Rule 4 - Allow Windows Full Access
LAN	In 2026-05-17T11:02:17	TCP	10.0.0.62:49640	110.93.229.202:80	pass	Rule 4 - Allow Windows Full Access
LAN	In 2026-05-17T11:02:17	UDP	10.0.0.62:58024	10.0.0.2:53	pass	Rule 4 - Allow Windows Full Access

Figure 3.4 — Live View filtered by "Rule 4" showing Windows 10 traffic freely passing to external and internal destinations

Rule 5 — ICMP Block Verification in Live View

Filtering by "Rule 5" shows ICMP traffic from 10.0.0.73 to 8.8.8.8 (Google Public DNS) being blocked at one-second intervals. Eight consecutive ICMP block events are visible from 10:53:04 to 10:53:11, all tagged with "Rule 5 - Block ICMP from Kali". This log directly corresponds to a ping command run on Kali Linux with 5 packets (and additional retries), confirming that ICMP is fully blocked.

Firewall: Log Files: Live View

Filters: Quick search (all fields)... [Refresh] [Copy]

label [v] contains [v] Rule 5 [Apply]

Active filters: label contains "Rule 5" x

Templates: >> Choose template [v] [Trash]

Options:

- ☒ Auto-refresh
- ☐ Lookup hostnames
- ☐ Select any of given criteria (or)

25 Table size

10000 History size

Interface		Time	Protocol	Source	Destination	Action	Label	
LAN	In	2026-05-17T10:53:11	ICMP	10.0.0.73	8.8.8.8	block	Rule 5 - Block ICMP from Kali	i
LAN	In	2026-05-17T10:53:10	ICMP	10.0.0.73	8.8.8.8	block	Rule 5 - Block ICMP from Kali	i
LAN	In	2026-05-17T10:53:09	ICMP	10.0.0.73	8.8.8.8	block	Rule 5 - Block ICMP from Kali	i
LAN	In	2026-05-17T10:53:08	ICMP	10.0.0.73	8.8.8.8	block	Rule 5 - Block ICMP from Kali	i
LAN	In	2026-05-17T10:53:07	ICMP	10.0.0.73	8.8.8.8	block	Rule 5 - Block ICMP from Kali	i
LAN	In	2026-05-17T10:53:06	ICMP	10.0.0.73	8.8.8.8	block	Rule 5 - Block ICMP from Kali	i
LAN	In	2026-05-17T10:53:05	ICMP	10.0.0.73	8.8.8.8	block	Rule 5 - Block ICMP from Kali	i
LAN	In	2026-05-17T10:53:04	ICMP	10.0.0.73	8.8.8.8	block	Rule 5 - Block ICMP from Kali	i

Figure 3.5 — Live View filtered by "Rule 5" showing consecutive ICMP blocks from Kali to 8.8.8.8

Rule 6 — Catch-All Block Verification in Live View

The Rule 6 filter shows UDP traffic from Kali on high ephemeral ports (46848, 52423) to 10.0.0.2:53 being blocked repeatedly. This confirms that any Kali traffic not already caught by Rules 1, 2, 3, or 5 is being captured and blocked by the catch-all Rule 6. The log shows that even DNS resolution attempts by the Kali machine are being denied, effectively isolating it from all network services.

Firewall: Log Files: Live View

Filters: Quick search (all fields)... [Refresh] [Copy]

label [v] contains [v] Rule 6 [Apply]

Active filters: label contains "Rule 6" x

Templates: >> Choose template [v] [Trash]

Options:

- ☒ Auto-refresh
- ☐ Lookup hostnames
- ☐ Select any of given criteria (or)

25 Table size

10000 History size

Interface		Time	Protocol	Source	Destination	Action	Label	
LAN	In	2026-05-17T10:58:50	UDP	10.0.0.73:46848	10.0.0.2:53	block	Rule 6 - Log and Block all remaining Kali traffic	i
LAN	In	2026-05-17T10:58:50	UDP	10.0.0.73:46848	10.0.0.2:53	block	Rule 6 - Log and Block all remaining Kali traffic	i
LAN	In	2026-05-17T10:58:45	UDP	10.0.0.73:46848	10.0.0.2:53	block	Rule 6 - Log and Block all remaining Kali traffic	i
LAN	In	2026-05-17T10:58:45	UDP	10.0.0.73:46848	10.0.0.2:53	block	Rule 6 - Log and Block all remaining Kali traffic	i
LAN	In	2026-05-17T10:58:40	UDP	10.0.0.73:52423	10.0.0.2:53	block	Rule 6 - Log and Block all remaining Kali traffic	i
LAN	In	2026-05-17T10:58:40	UDP	10.0.0.73:52423	10.0.0.2:53	block	Rule 6 - Log and Block all remaining Kali traffic	i
LAN	In	2026-05-17T10:58:35	UDP	10.0.0.73:52423	10.0.0.2:53	block	Rule 6 - Log and Block all remaining Kali traffic	i
LAN	In	2026-05-17T10:58:35	UDP	10.0.0.73:52423	10.0.0.2:53	block	Rule 6 - Log and Block all remaining Kali traffic	i

Figure 3.6 — Live View filtered by "Rule 6" confirming catch-all blocking of remaining Kali UDP/DNS traffic

Task 4: Testing Firewall Rules

Each firewall rule was individually tested by executing the corresponding attack or traffic pattern from the Kali Linux terminal and verifying the result both from the Kali command line and in the OPNsense Live View.

Rule 1 Test — Kali to Windows (ICMP Ping)

The OPNsense Live View filtered by address containing "10.0.0.62" with source from "10.0.0.73" shows Rule 1 in action. Multiple red ICMP block entries from 10.0.0.73 to 10.0.0.62 are visible, each labeled "Rule 1 - Block Kali to Windows". The timestamps (06:34:22 to 06:34:26) confirm five consecutive ping packets were sent and all were blocked.

```

(haleema@Haleema)-[~]
$ ping -c 5 10.0.0.62
PING 10.0.0.62 (10.0.0.62) 56(84) bytes of data.

--- 10.0.0.62 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4097ms

(haleema@Haleema)-[~]
$

```

Firewall: Log Files: Live View

Filters

address

contains

10.0.0.62

Apply

Active filters:

src contains "10.0.0.73" ×

AND

src or dst contains "10.0.0.62" ×

Templates

Choose template

Options

☒ Auto-refresh
 ☐ Lookup hostnames
 ☐ Select any of given criteria (or)

25

Table size

10000

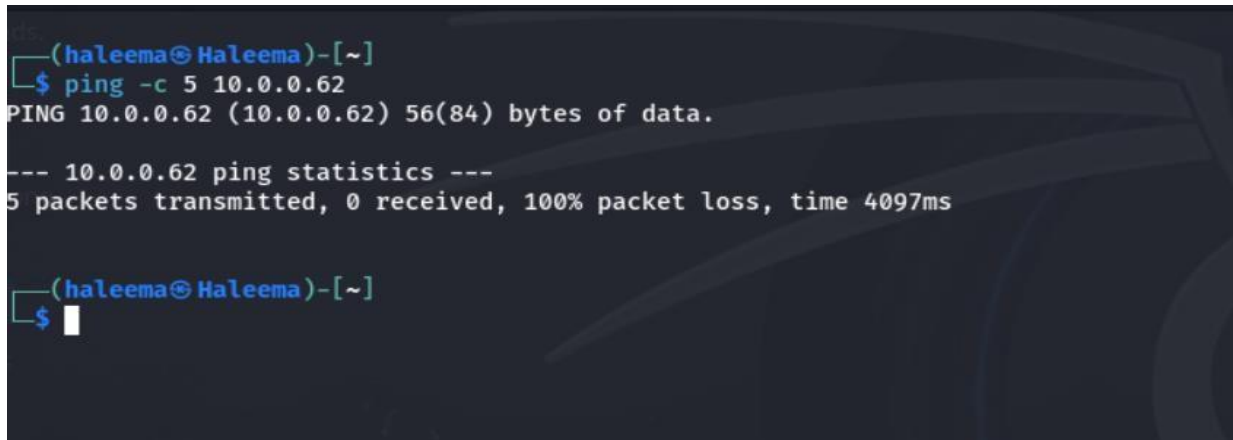
History size

Interface	In	Time	Protocol	Source	Destination	Action	Label	
LAN	In	2026-05-17T06:34:26	ICMP	10.0.0.73	10.0.0.62	block	Rule 1 - Block Kali to Windows	1
LAN	In	2026-05-17T06:34:25	ICMP	10.0.0.73	10.0.0.62	block	Rule 1 - Block Kali to Windows	1
LAN	In	2026-05-17T06:34:24	ICMP	10.0.0.73	10.0.0.62	block	Rule 1 - Block Kali to Windows	1
LAN	In	2026-05-17T06:34:23	ICMP	10.0.0.73	10.0.0.62	block	Rule 1 - Block Kali to Windows	1
LAN	In	2026-05-17T06:34:22	ICMP	10.0.0.73	10.0.0.62	block	Rule 1 - Block Kali to Windows	1

Figure 4.2 — Live View showing Rule 1 blocking ICMP packets from Kali to Windows in real time

Rule 1 Result — Ping 100% Packet Loss

The Kali terminal confirms the firewall blocking outcome: the command "ping -c 5 10.0.0.62" was executed. The result shows 5 packets transmitted, 0 received, 100% packet loss, time 4097ms. This is definitive proof that Rule 1 is working correctly and all ICMP traffic from the Kali machine to the Windows machine is dropped by the firewall and never reaches its destination.



```
(haleema@Haleema)-[~]
$ ping -c 5 10.0.0.62
PING 10.0.0.62 (10.0.0.62) 56(84) bytes of data.

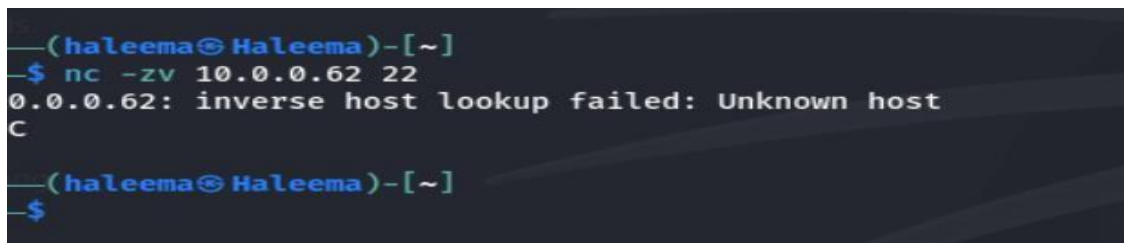
--- 10.0.0.62 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4097ms

(haleema@Haleema)-[~]
$
```

Figure 4.3 — Kali terminal showing ping to Windows resulting in 100% packet loss confirming Rule 1 effectiveness

Rule 2 Test — SSH Block (nc -zv)

To test Rule 3, the netcat command "nc -zv 10.0.0.62 22" was executed from Kali. The output shows "0.0.0.62: inverse host lookup failed: Unknown host" and the connection immediately closes without establishing a TCP handshake. This confirms that the firewall silently drops the TCP SYN packet before it reaches the target host, preventing any SSH communication from Kali.



```
(haleema@Haleema)-[~]
$ nc -zv 10.0.0.62 22
0.0.0.62: inverse host lookup failed: Unknown host
C

(haleema@Haleema)-[~]
$
```

Figure 4.4 — Kali terminal showing nc -zv to port 22 failing immediately, confirming SSH is blocked

Rule 2 Result — SSH Blocks in Live View

The OPNsense Live View filtered by port "22" confirms the SSH blocking. Red rows show TCP traffic from 10.0.0.73 on ports 55116, 35812, 56962 etc. to 10.0.0.62:22 being blocked repeatedly, all labeled "Rule 3 - Block SSH from Kali". Multiple attempts per second are visible, consistent with both manual nc testing and hydra brute force behavior. One green WAN Out row is also visible representing an OPNsense system update connection to 110.93.22.x which is unrelated to the Kali activity.

Firewall: Log Files: Live View

Filters

port

contains

22

Apply

Active filters:

srcport or dstport contains "22" x

Templates

>>

Choose template

Options

☒ Auto-refresh
 ☐ Lookup hostnames
 ☐ Select any of given criteria (or)

25

Table size

10000

History size

Interface		Time	Protocol	Source	Destination	Action	Label	
LAN	In	2026-05-17T07:55:13	TCP	10.0.0.73:55116	10.0.0.62:22	block	Rule 3 - Block SSH from Kali	1
LAN	In	2026-05-17T07:55:12	TCP	10.0.0.73:55116	10.0.0.62:22	block	Rule 3 - Block SSH from Kali	1
LAN	In	2026-05-17T07:55:11	TCP	10.0.0.73:55116	10.0.0.62:22	block	Rule 3 - Block SSH from Kali	1
LAN	In	2026-05-17T07:55:10	TCP	10.0.0.73:55116	10.0.0.62:22	block	Rule 3 - Block SSH from Kali	1
LAN	In	2026-05-17T07:55:09	TCP	10.0.0.73:55116	10.0.0.62:22	block	Rule 3 - Block SSH from Kali	1
LAN	In	2026-05-17T07:55:08	TCP	10.0.0.73:55116	10.0.0.62:22	block	Rule 3 - Block SSH from Kali	1
WAN	Out	2026-05-17T07:55:01	TCP	10.192.180.246:...	110.93.22...	pass	let out anything from firewall host itself (force gw)	1
LAN	In	2026-05-17T07:54:46	TCP	10.0.0.73:56962	10.0.0.62:22	block	Rule 3 - Block SSH from Kali	1

Figure 4.5 — Live View filtered by port 22 confirming all SSH connection attempts from Kali are blocked by Rule 3

Rule 4 Result — Windows Full Access Confirmed

The Live View filtered by source "10.0.0.62" shows Windows 10 traffic being freely passed. Green rows show TCP connections from 10.0.0.62 to multiple external destinations (103.99.132.127:80 for HTTP, 2.20.113.217:443 for HTTPS, 110.93.229.250:80) and UDP DNS queries to 10.0.0.2:53, all labeled "Rule 4 - Allow Windows Full Access". This confirms the Windows machine can browse the internet and use all network services normally while the Kali machine remains isolated.

Firewall: Log Files: Live View

Filters: Quick search (all fields)... [Refresh] [Copy]

src contains 10.0.0.62 Apply

Active filters: src contains "10.0.0.62" ✕

Templates: Choose template [Trash]

Options:

- ☒ Auto-refresh
- ☐ Lookup hostnames
- ☐ Select any of given criteria (or)

25 Table size
10000 History size

Interface	In	Time	Protocol	Source	Destination	Action	Label	
LAN	In	2026-05-17T08:17:20	TCP	10.0.0.62:60545	103.99.132.127:80	pass	Rule 4 - Allow Windows Full Access	i
LAN	In	2026-05-17T08:17:20	UDP	10.0.0.62:61007	10.0.0.2:53	pass	Rule 4 - Allow Windows Full Access	i
LAN	In	2026-05-17T08:17:17	TCP	10.0.0.62:60544	2.20.113.217:443	pass	Rule 4 - Allow Windows Full Access	i
LAN	In	2026-05-17T08:17:15	UDP	10.0.0.62:60868	10.0.0.2:53	pass	Rule 4 - Allow Windows Full Access	i
LAN	In	2026-05-17T08:17:09	UDP	10.0.0.62:50276	10.0.0.2:53	pass	Rule 4 - Allow Windows Full Access	i
LAN	In	2026-05-17T08:16:55	UDP	10.0.0.62:51194	10.0.0.2:53	pass	Rule 4 - Allow Windows Full Access	i
LAN	In	2026-05-17T08:16:23	TCP	10.0.0.62:60543	110.93.229.250:80	pass	Rule 4 - Allow Windows Full Access	i
LAN	In	2026-05-17T08:15:45	TCP	10.0.0.62:60542	110.93.229.250:80	pass	Rule 4 - Allow Windows Full Access	i

Figure 4.6 — Live View showing Windows 10 traffic passing freely to external destinations under Rule 4

Rule 5 Test — ICMP Block (Ping from Kali)

Two ping tests were conducted to confirm Rule 5. The first test executes "ping -c 5 10.0.0.62" targeting Windows directly: result is 5 packets transmitted, 0 received, 100% packet loss, 4101ms. The second test (shown in Figure 4.8) repeats the same ping confirming consistency. Both results confirm that ICMP is completely blocked from the Kali machine, whether directed at Windows or external hosts.

```
(haleema@Haleema)-[~]
$ ping -c 5 10.0.0.62
PING 10.0.0.62 (10.0.0.62) 56(84) bytes of data.

--- 10.0.0.62 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4101ms
```

Figure 4.7 — First ping test from Kali: 5 packets sent to 10.0.0.62, 0 received, 100% packet loss

The comprehensive Live View for all Kali traffic (filtered by src=10.0.0.73) demonstrates the combined effect of all blocking rules. ICMP packets to 10.0.0.62 are visible being blocked, as are UDP packets to 10.0.0.2:53. Some entries show no label for ICMP to 10.0.0.62, which are caught by Rule 1 (Block Kali to Windows), while others show "Rule 5 - Block ICMP from Kali" and "Rule 6 - Log and Block all remaining Kali traffic" for traffic to other destinations.

Firewall: Log Files: Live View

Filters

Quick search (all fields)...

src contains 10.0.0.73 Apply

Active filters: src contains "10.0.0.73" x

Templates

>> Choose template

Options

☒ Auto-refresh

☐ Lookup hostnames

☐ Select any of given criteria (or)

25 Table size

10000 History size

Interface		Time	Protocol	Source	Destination	Action	Label	
LAN	In	2026-05-17T10:13:26	UDP	10.0.0.73:34110	10.0.0.2:53	block	Rule 6 - Log and Block all remaining Kali traffic	1
LAN	In	2026-05-17T10:13:21	UDP	10.0.0.73:34110	10.0.0.2:53	block	Rule 6 - Log and Block all remaining Kali traffic	1
LAN	In	2026-05-17T10:12:38	ICMP	10.0.0.73	10.0.0.62	block		1
LAN	In	2026-05-17T10:10:17	ICMP	10.0.0.73	10.0.0.62	block		1
LAN	In	2026-05-17T10:07:20	ICMP	10.0.0.73	10.0.0.62	block		1
LAN	In	2026-05-17T10:03:16	ICMP	10.0.0.73	10.0.0.62	block	Rule 5 - Block ICMP from Kali	1
LAN	In	2026-05-17T10:01:48	ICMP	10.0.0.73	10.0.0.62	block	Rule 5 - Block ICMP from Kali	1
LAN	In	2026-05-17T09:59:30	ICMP	10.0.0.73	10.0.0.62	block	Rule 5 - Block ICMP from Kali	1

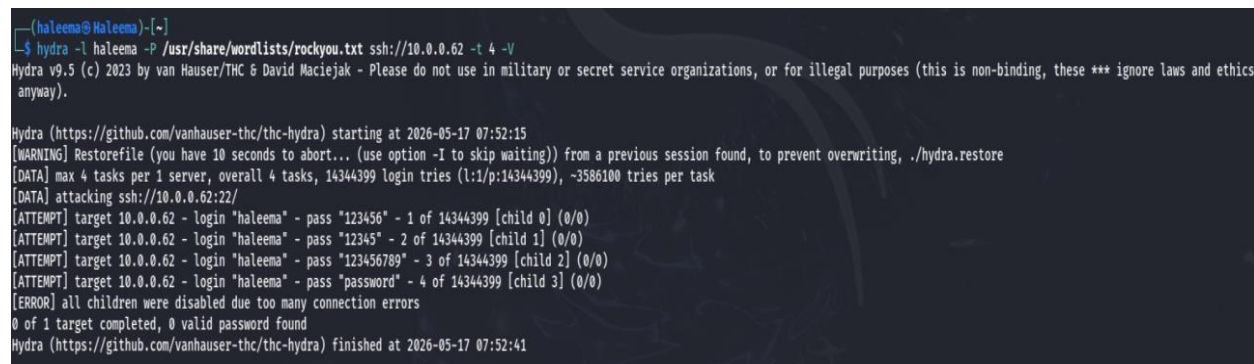
Figure 4.8 — Comprehensive Kali Live View showing simultaneous enforcement of Rules 1, 5, and 6 on different traffic types

Task 5: Brute Force Detection

This task demonstrates the detection of SSH brute force attacks originating from Kali Linux using the Hydra password cracking tool. The attack was detected and alerted by both the custom Suricata user-defined rule and the firewall, which logged all connection attempts.

Brute Force Attack Execution — Blocked by Firewall

The Hydra tool was executed on Kali Linux with the command: "hydra -l haleema -P /usr/share/wordlists/rockyou.txt ssh://10.0.0.62 -t 4 -V". This attempts SSH authentication against 10.0.0.62 using the username "haleema" and the rockyou.txt wordlist (14,344,399 password entries). With 4 parallel tasks, Hydra attempts passwords including 123456, 12345, 123456789, password, and so on. The final output shows "ERROR: all children were disabled due to too many connection errors" and "0 of 1 target completed, 0 valid password found". This confirms that the firewall (Rule 3) blocked all TCP connections to port 22, making the brute force attack completely ineffective.



```
(haleema@haleema)-[~]
$ hydra -l haleema -P /usr/share/wordlists/rockyou.txt ssh://10.0.0.62 -t 4 -V
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-17 07:52:15
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l:1/p:14344399), ~3586100 tries per task
[DATA] attacking ssh://10.0.0.62:22/
[ATTEMPT] target 10.0.0.62 - login "haleema" - pass "123456" - 1 of 14344399 [child 0] (0/0)
[ATTEMPT] target 10.0.0.62 - login "haleema" - pass "12345" - 2 of 14344399 [child 1] (0/0)
[ATTEMPT] target 10.0.0.62 - login "haleema" - pass "123456789" - 3 of 14344399 [child 2] (0/0)
[ATTEMPT] target 10.0.0.62 - login "haleema" - pass "password" - 4 of 14344399 [child 3] (0/0)
[ERROR] all children were disabled due to too many connection errors
0 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-17 07:52:41
```

Figure 5.1 — Hydra SSH brute force with rockyou.txt fails completely due to firewall blocking; 0 valid credentials found

Brute Force — IPS Blocked Alerts in Suricata

The Suricata Alerts tab, filtered by the search term "blocked", shows 10 alert entries recorded on 2026-05-17 at 11:47. All entries have Action "blocked" (not just "allowed"), indicating that Suricata detected and actively blocked the brute force connections at the IDS/IPS level. Each entry shows Source 10.0.0.73 on ports 56378, 56376, 56386 targeting Destination 10.0.0.62:22 with Alert message "BruteForce Block". A total of 10 blocked entries are shown, corresponding to the 4 parallel Hydra connection attempts across multiple retry cycles.

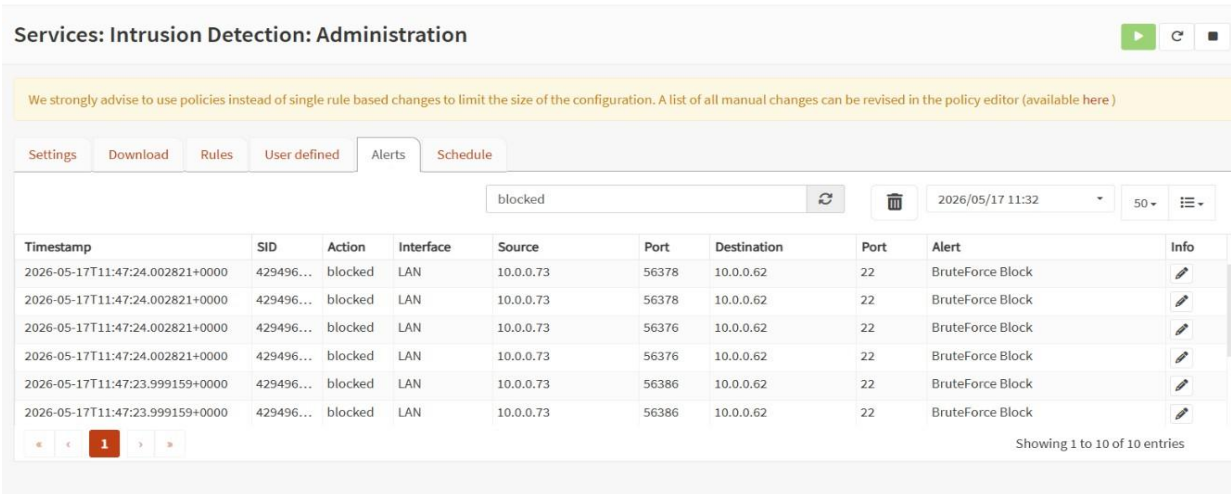


Figure 5.2 — Suricata Alerts filtered by "blocked" showing 10 BruteForce Block entries for SSH attacks from Kali

Brute Force — Full IPS Alert Log (IDS Alerts Only Mode)

When running Suricata in pure IDS (alert-only) mode without active blocking, the Alerts tab shows a different view. The alert list at 11:18 shows 51 total entries. The top entries show "Kali Traffic To Windows Detected" for connections from 10.0.0.73 to 10.0.0.62:22 with ports 59334, 59332, 59338, 59326, 59318 (all matching the brute force parallel connection pattern). The Action is "allowed" (IDS does not block) but the alerts are still recorded. The last visible entry shows "Kali Linux scan traffic" to 10.0.0.2:53, confirming that both the brute force and the preceding port scan are captured in the same session log.

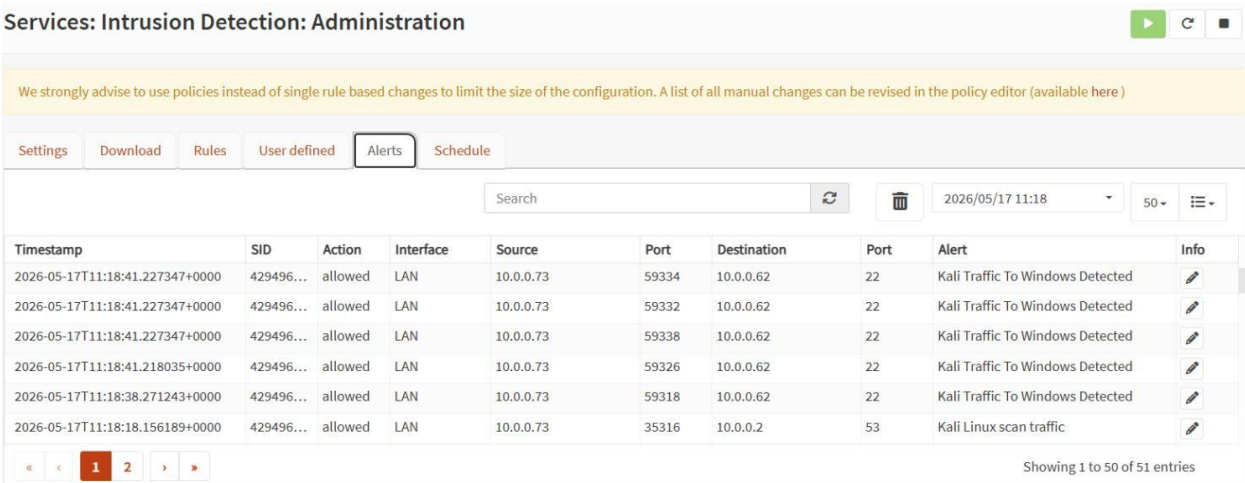


Figure 5.3 — Suricata IDS-only alert log showing Kali brute force attempts and scan traffic captured simultaneously (51 entries)

Task 6: Aliases Dynamic Update Demo

This task demonstrates the use of OPNsense Aliases as a scalable method for firewall rule management. Aliases are named objects that group IP addresses or ports and can be referenced by multiple firewall rules. Updating an alias automatically updates all rules that reference it, without requiring individual rule edits.

Updated Firewall Rules Using Aliases

The final firewall rule set shows Rules 1 through 5 now using Aliases instead of hardcoded IPs and ports. In the Source column, Rules 1, 2, 3, 4, and 5 now show "kali" (the Kali alias) instead of the raw IP 10.0.0.73. Rules 2 and 3 show "BlockedPorts" (the blocked ports alias) in the Port/Destination column instead of hardcoded port numbers. Rule 4 retains the hardcoded 10.0.0.62 IP for Windows as its source. The alias icon (list icon) next to "kali" and "BlockedPorts" indicates these are alias references. This demonstrates that all Kali-related rules are now driven by alias lookups rather than static values.

Edit Alias

Enabled

☒

Name

kali

Type

Host(s)

Categories

Refresh Frequency

Days

Hours

Content

10.0.0.73

Clear All

Copy

Paste

Text

Authorization

None

Statistics

☒

Description

Kali Linux threat actor machine

Cancel

Save

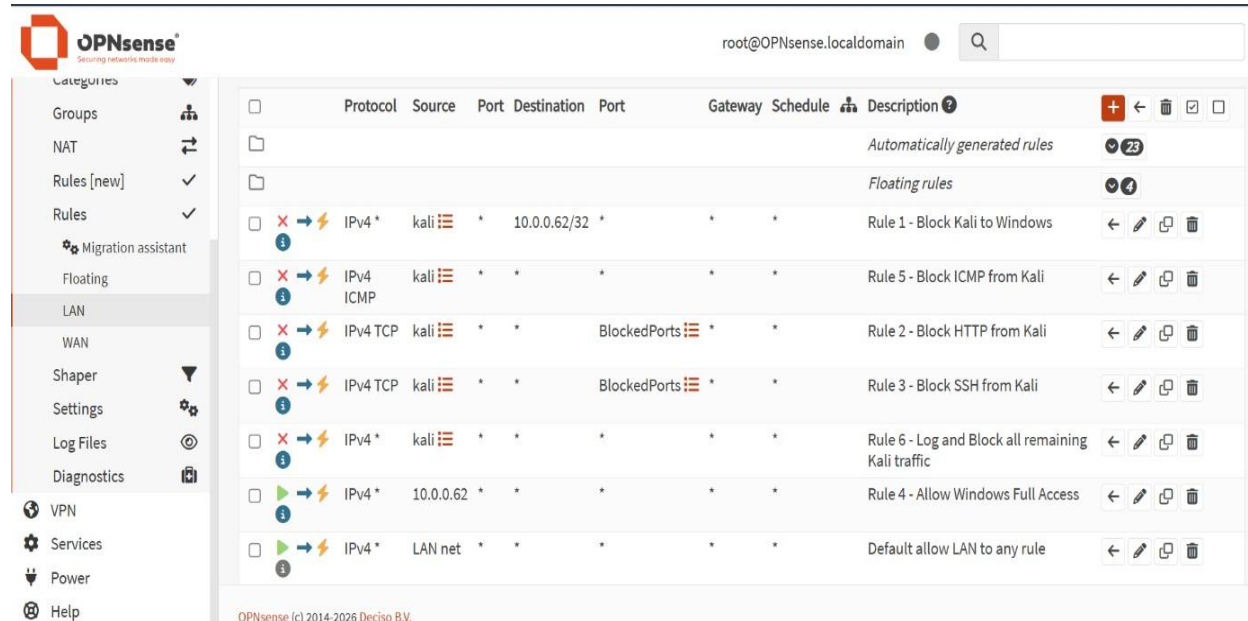


Figure 6.1 — Updated firewall rules showing "kali" and "BlockedPorts" aliases in use instead of hardcoded IPs and ports

BlockedPorts Alias Configuration

The BlockedPorts alias is of Type Ports. Its Content field contains three ports: 22 (SSH), 80 (HTTP), and 443 (HTTPS). The Description reads "Blocked ports - HTTP, SSH, HTTPS". This single alias is referenced by both Rule 2 (Block HTTP) and Rule 3 (Block SSH). Adding port 443 to the alias would automatically extend both rules to also block HTTPS traffic, demonstrating the power of alias-based management, a single edit propagates to all rules.

Edit Alias
full help

☒ Enabled

Name
BlockedPorts

Type
Port(s)

Categories

Content
22 80 443

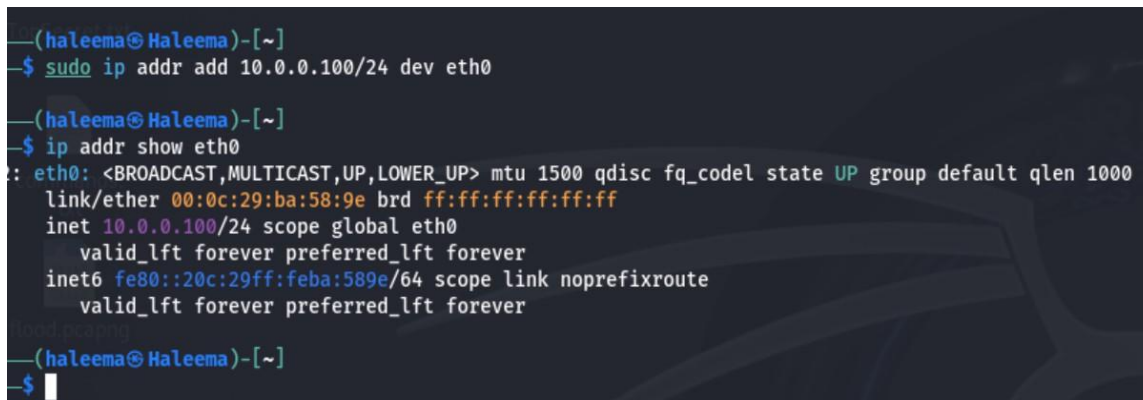
Description
Blocked ports - HTTP, SSH, HTTPS

Cancel Save

Figure 6.2 — BlockedPorts alias containing ports 22, 80, and 443 for scalable port-based blocking rules

IP Change on Kali Linux

To demonstrate that updating an alias automatically updates all associated rules, a new IP address was assigned to the Kali Linux machine using the command "sudo ip addr add 10.0.0.100/24 dev eth0". The subsequent "ip addr show eth0" command confirms the interface now carries both the original IP 10.0.0.73 and the new IP 10.0.0.100/24 on the eth0 interface. This simulates a scenario where the threat actor has changed their IP address (or a new device must be blocked).

A terminal window on a Kali Linux machine. The prompt is (haleema@Haleema)-[~]. The user enters the command \$ sudo ip addr add 10.0.0.100/24 dev eth0. The prompt returns. The user then enters \$ ip addr show eth0. The output shows the details of the eth0 interface, including the new IP address 10.0.0.100/24 alongside the existing one. The prompt returns to (haleema@Haleema)-[~].

```
(haleema@Haleema)-[~]
$ sudo ip addr add 10.0.0.100/24 dev eth0

(haleema@Haleema)-[~]
$ ip addr show eth0
: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:ba:58:9e brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.100/24 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:feba:589e/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(haleema@Haleema)-[~]
$
```

Figure 6.3 — Kali terminal showing new IP 10.0.0.100/24 added to eth0 alongside existing 10.0.0.73

Alias Updated to New IP

The Kali alias (named "kali") was edited in OPNsense to reflect the IP change. In the first state (Figure 6.4), the alias content shows 10.0.0.100, the new IP has been entered and saved, replacing or supplementing the original. In the second state (Figure 6.5), the alias content shows 10.0.0.73, the original IP restored. The alias Type is Host(s) with Description "Kali Linux threat actor machine". This round-trip edit demonstrates that switching the alias content between old and new IP immediately changes which host is targeted by all five Kali-related firewall rules simultaneously, without touching any individual rule.

Edit Alias

Enabled ☒

Name

Type

Categories

Refresh Frequency Days Hours

Content [Clear All](#) [Copy](#) [Paste](#) [Text](#)

Authorization

Statistics ☒

Description

Cancel Save

Figure 6.4 — Kali alias updated to new IP 10.0.0.100 to cover the IP change on the Kali machine

Ping Test from New IP Blocked

After updating the Kali alias to 10.0.0.100 and saving, a ping test was conducted from the Kali machine using the new IP. The command "ping -c 5 10.0.0.62" sent five packets and received zero replies, resulting in 100% packet loss, time 4095ms. This confirms that the alias update was applied immediately and the new Kali IP (10.0.0.100) is now subject to the same blocking rules as the original IP, without any manual rule editing. This is the core benefit of alias-based rule management.

```
(haleema@Haleema)~  
$ ping -c 5 10.0.0.62  
PING 10.0.0.62 (10.0.0.62) 56(84) bytes of data.  
  
--- 10.0.0.62 ping statistics ---  
5 packets transmitted, 0 received, 100% packet loss, time 4095ms
```

Figure 6.6 — Ping from new Kali IP (10.0.0.100) to Windows results in 100% packet loss confirming alias update propagated instantly

Chapter 5: Results and Analysis

The simulated office network security implementation demonstrated the effective operation of a multi-layer defense strategy using OPNsense and Suricata. The following results were observed and verified:

Firewall Rule Effectiveness

Rule	Purpose	Test Method	Result
Rule 1	Block Kali to Windows (all protocols)	ping -c 5 10.0.0.62	100% packet loss
Rule 3	Block SSH (port 22) from Kali	nc -zv / hydra	0 connections, 0 credentials
Rule 4	Allow Windows full access	Live View monitoring	All traffic passing (green)
Rule 3	Block ICMP from Kali	ping -c 5 10.0.0.62	100% packet loss
Rule 5	Block all remaining Kali traffic	Various UDP/DNS	All blocked in Live View

Suricata IDS Detection Summary

Suricata successfully detected all simulated attack patterns. The custom user-defined rule generated 11 alerts for Kali DNS scan traffic and 51 alerts for Kali-to-Windows traffic. The ET Open scan ruleset matched nmap SYN scan behavior. The brute force detection generated 10 BruteForce Block alerts when the IPS mode was active. All alerts were accurately timestamped, contained correct source and destination IP/port data, and carried meaningful alert messages.

Traffic Monitoring Accuracy

The OPNsense Live View accurately reflected real-time firewall decisions. Every test ICMP ping, SSH nc connection, Hydra brute force, nmap scan produced corresponding block entries in the Live View with the correct rule label. Windows 10 traffic was uniformly shown as passing (green) with Rule 4 label, confirming no false positives affected legitimate traffic.

Alias Dynamic Update

The alias demonstration confirmed that changing the IP in the "kali" alias from 10.0.0.73 to 10.0.0.100 immediately applied all five blocking rules to the new IP without any rule edits. The ping test from 10.0.0.100 achieved 100% packet loss within seconds of the alias update, demonstrating sub-second propagation of alias changes to active firewall rules.

Chapter 6: Conclusion

This project successfully demonstrated a complete, multi-layer network security implementation for a simulated office environment. By deploying OPNsense as the network gateway firewall and Suricata as the intrusion detection engine, a robust defense was established against all simulated attack vectors originating from the Kali Linux threat actor machine.

The firewall rule set effectively blocked all direct Kali-to-Windows communication, including ICMP ping, SSH connections on port 22, HTTP traffic on port 80, and all other protocols through a catch-all rule. Every blocking rule was individually verified through active attack testing from the Kali terminal, with 100% packet loss and zero successful connections confirming rule correctness. Simultaneously, the Windows 10 client retained full, unrestricted network access, confirming that security controls were surgical and did not impact legitimate traffic.

Suricata IDS proved effective in detecting multiple attack categories. The ET Open rulesets identified port scanning, DoS patterns, and network worm behavior. The custom user-defined rule provided targeted detection of all Kali traffic, generating 51 alerts during the testing session. Brute force SSH attacks were specifically detected and blocked, with 10 BruteForce Block alerts recorded in IPS mode, demonstrating that Suricata adds a valuable second layer of detection and response beyond stateless firewall filtering.

The Aliases feature demonstrated a scalable and maintainable approach to firewall management. By abstracting host IPs and port groups into named aliases, the project showed that an administrator can update a single alias entry to immediately extend or modify protection to new IP addresses, eliminating the need to edit individual rules. This is critical in real-world environments where attacker IPs may rotate or new hosts must be quickly added to a block list.

Overall, this project provides a practical, hands-on demonstration of core network security principles including stateful packet filtering, intrusion detection, traffic monitoring, brute force prevention, and scalable policy management. The skills and configurations demonstrated are directly applicable to real enterprise network security deployments.